Optimization

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Lesson 3 Exercises

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Exercise 2

Determine the maximum and the minimum absolute points of the function

$$\begin{aligned} f(x,y) &= x^2 + 2y^2 \\ \text{s.t.} \quad & x^2 + y^2 \leq 1 \\ & x + y - 1 = 0 \end{aligned}$$

applying the KKT conditions.

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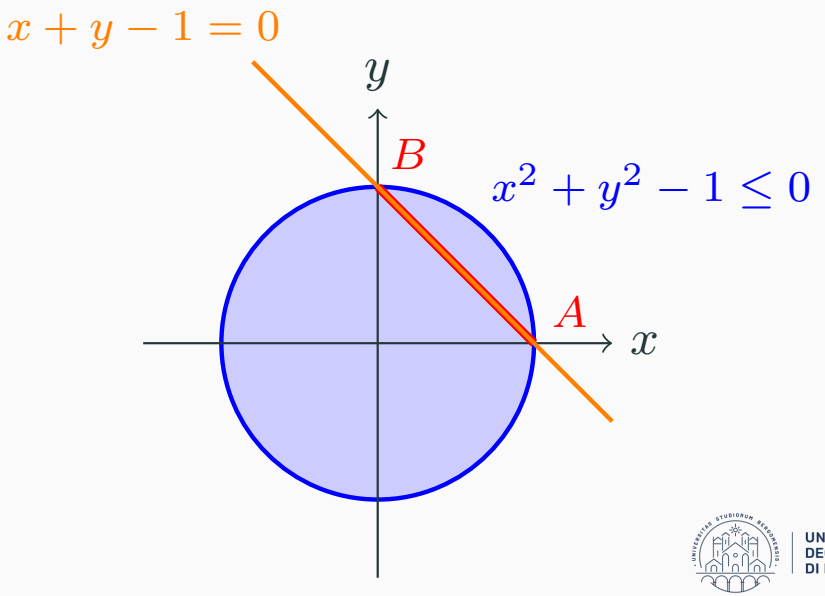
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$$f(x,y) = x^2 + 2y^2$$
$$\text{s.t.} \quad x^2 + y^2 \leq 1$$
$$x + y - 1 = 0$$

applying the KKT conditions.

Solution: First, we observe that the **feasible region** is a **closed and bounded set**. Therefore, because of the Weierstrass theorem, f admits an absolute maximum and an absolute minimum on the feasible region.



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Exercise 2

- We should notice that on all the points on the segment of the feasible region, with the exception of the extreme points A and B, there is only one active constraint (that is $x + y - 1 = 0$) and, consequently, all these points are regular. We need to check if A and B are regular points (the gradient of the constraints in that points are linearly independent).
- Let $g_1(x, y) = x^2 + y^2 - 1 \leq 0$ and $g_2(x, y) = x + y - 1 = 0$. Then

$$\nabla g_1(x, y) = \begin{bmatrix} 2x \\ 2y \end{bmatrix} \quad \nabla g_2(x, y) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\nabla g_1(A) = \begin{bmatrix} 2 \\ 0 \end{bmatrix} \quad \nabla g_2(A) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \nabla g_1(B) = \begin{bmatrix} 0 \\ 2 \end{bmatrix} \quad \nabla g_2(B) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

It follows that

$$\det \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} = 2 \neq 0 \quad \det \begin{bmatrix} 0 & 1 \\ 2 & 1 \end{bmatrix} = -2 \neq 0$$

which means that **the regularity conditions are satisfied also for A and B.**

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Exercise 2

- We now write the Lagrangian function, that is

$$\mathcal{L}(x, y, \lambda_1, \lambda_2) = x^2 + 2y^2 - \lambda_1(x^2 + y^2 - 1) - \lambda_2(x + y - 1)$$

and we derive the following KKT conditions

$$\begin{cases} \frac{\partial \mathcal{L}}{\partial x} = 0 \\ \frac{\partial \mathcal{L}}{\partial y} = 0 \\ x^2 + y^2 - 1 \leq 0 \\ \lambda_1(x^2 + y^2 - 1) = 0, \quad \lambda_1 \geq 0 \\ x + y - 1 = 0 \end{cases} \Longrightarrow \begin{cases} 2x - 2\lambda_1 x - \lambda_2 = 0 \\ 4y - 2\lambda_1 y - \lambda_2 = 0 \\ x^2 + y^2 - 1 \leq 0 \\ \lambda_1(x^2 + y^2 - 1) = 0, \quad \lambda_1 \geq 0. \\ x + y - 1 = 0 \end{cases}$$

- Starting from the complementarity condition $\lambda_1(x^2 + y^2 - 1) = 0$, it follows that $\lambda_1 = 0$ or $x^2 + y^2 - 1 = 0$. The solutions of the system are them the following:

$$\mathbf{x}^{(0)} = \begin{bmatrix} 2/3 \\ 1/3 \end{bmatrix} \text{ with } \boldsymbol{\lambda}^{(0)} = \begin{bmatrix} 0 \\ 4/3 \end{bmatrix}, \ \mathbf{x}^{(1)} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \text{ with } \boldsymbol{\lambda}^{(1)} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$\mathbf{x}^{(2)} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \text{ with } \boldsymbol{\lambda}^{(2)} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

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Exercise 2

- If we now compute the value of the objective function f in the three points $\mathbf{x}^{(0)}, \mathbf{x}^{(1)}$, and $\mathbf{x}^{(2)}$, then

$$f(\mathbf{x}^{(0)}) = \frac{2}{3} \quad f(\mathbf{x}^{(1)}) = 1 \quad f(\mathbf{x}^{(2)}) = 2.$$

- We can conclude that $\mathbf{x}^{(0)}$ is the **global minimum** and $\mathbf{x}^{(2)}$ is the **global maximum**.

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Exercise 3

Determine the minimum of the function

$$\begin{aligned} \min f(x,y) &= -x - y^2 \\ \text{s.t.} \quad & 2x^2 + y^2 \leq 7 \\ & x^2 + y^2 \geq 2 \end{aligned}$$

applying the KKT conditions.

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Solution:

- First, the constraints should be written as:

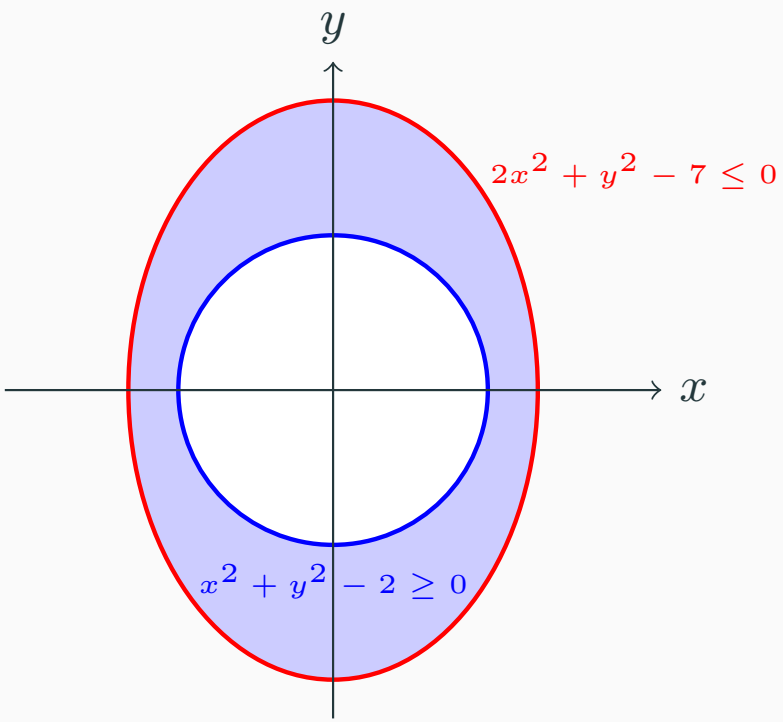
$$\begin{aligned} g_1(x,y) &= 2x^2 + y^2 - 7 \leq 0 \\ g_2(x,y) &= -x^2 - y^2 + 2 \leq 0, \end{aligned}$$

  according to the standard formulation of a minimization problem.

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Exercise 3

- The **feasible region** consists of the intersection between the interior of the ellipse defined by $g_1(x, y)$ and the exterior of the circle defined by $g_2(x, y)$, as illustrated below.



We note that there is no point on which both constraints are active, as the two curves have no intersections. In addition, the gradient of each constraint is zero only in $(0, 0)$, that does not belong to the feasible regions. Then, **the regularity conditions are always satisfied**.

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- We now write the Lagrangian function, that is

$$\mathcal{L}(x, y, \lambda_1, \lambda_2) = -x - y^2 + \lambda_1(2x^2 + y^2 - 7) + \lambda_2(-x^2 - y^2 + 2)$$

and we derive the following KKT conditions

$$\begin{cases} \frac{\partial \mathcal{L}}{\partial x} = 0 \\ \frac{\partial \mathcal{L}}{\partial y} = 0 \\ 2x^2 + y^2 - 7 \leq 0 \\ \lambda_1(2x^2 + y^2 - 7) = 0, \quad \lambda_1 \geq 0 \\ -x^2 - y^2 + 2 \leq 0 \\ \lambda_2(-x^2 - y^2 + 2) = 0, \quad \lambda_2 \geq 0 \end{cases} \implies \begin{cases} -1 + 4\lambda_1 x - 2\lambda_2 x = 0 \\ -2y + 2\lambda_1 y - 2\lambda_2 y = 0 \\ 2x^2 + y^2 - 7 \leq 0 \\ \lambda_1(2x^2 + y^2 - 7) = 0, \quad \lambda_1 \geq 0 \\ -x^2 - y^2 + 2 \leq 0 \\ \lambda_2(-x^2 - y^2 + 2) = 0, \quad \lambda_2 \geq 0 \end{cases}$$

- If **both constraints are not active**, that is $\lambda_1 = \lambda_2 = 0$, then from the first KKT condition it follows that $-1 = 0$, which is impossible.
- If **both constraints are active**, then there are no feasible points as the two curves have no intersections.

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- Only $g_1(x, y)$ is active, that is $\lambda_2 = 0$. We first note that, since $\nabla g_1(x, y) = \begin{bmatrix} 4x \\ 2y \end{bmatrix}$ is equal to $\mathbf{0}$ only in $(0, 0)$, all the feasible points satisfy the regularity condition. Moving now to the KKT conditions, from $\lambda_2 = 0$ it follows that

$$\begin{cases} -1 + 4\lambda_1 x = 0 \\ -2y + 2\lambda_1 y = 0 \\ 2x^2 + y^2 - 7 = 0 \\ \lambda_1 \geq 0 \end{cases} \implies -2y(1 - \lambda_1) = 0 \implies y = 0 \ \vee \ y \neq 0$$

- If $y = 0$, then from $2x^2 + y^2 - 7 = 0$ we get the points $A\left(\sqrt{\frac{7}{2}}, 0\right)$ and $B\left(-\sqrt{\frac{7}{2}}, 0\right)$. However, from $-1 + 4\lambda_1 x = 0$ it follows that $\lambda_1 \geq 0$ only for A.
- If $y \neq 0$, then $\lambda_1 = 1$, and then $x = \frac{1}{4}$ from the first condition, while $y = \pm\sqrt{\frac{55}{8}}$ from the third condition. Hence, other two points are $C\left(\frac{1}{4}, \sqrt{\frac{55}{8}}\right)$ and $D\left(\frac{1}{4}, -\sqrt{\frac{55}{8}}\right)$

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- Only $g_2(x, y)$ is active, that is $\lambda_1 = 0$. As before, we note that, since $\nabla g_2(x, y) = \begin{bmatrix} -2x \\ -2y \end{bmatrix}$ is equal to $\mathbf{0}$ only in $(0, 0)$, all the feasible points satisfy the regularity condition. Moving now to the KKT conditions, from $\lambda_1 = 0$ it follows that

$$\begin{cases} -1 - 2\lambda_2 x = 0 \\ -2y - 2\lambda_2 y = 0 \\ -x^2 - y^2 + 2 = 0 \\ \lambda_2 \geq 0 \end{cases} \implies -2y(1 + \lambda_2) = 0 \implies y = 0 \ \vee \ y \neq 0$$

- If $y = 0$, then from $-x^2 - y^2 + 2 = 0$ we get the points $E\left(\sqrt{2}, 0\right)$ and $F\left(-\sqrt{2}, 0\right)$. However, from $-1 - 2\lambda_2 x = 0$ it follows that $\lambda_2 \geq 0$ only for E.
- If $y \neq 0$, then $\lambda_2 = -1$, in contrast with non negativity condition.

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- Therefore, the points that satisfy the necessary conditions are $A\left(\sqrt{\frac{7}{2}}, 0\right)$, $C\left(\frac{1}{4}, \sqrt{\frac{55}{8}}\right)$, $D\left(\frac{1}{4}, -\sqrt{\frac{55}{8}}\right)$, and $E\left(\sqrt{2}, 0\right)$.

- If evaluate the objective function in these points, it follows that

$$f(A) = -\sqrt{\frac{7}{2}}, \quad f(C) = f(D) = -\frac{57}{8}, \quad f(E) = \sqrt{2}.$$

  Since the feasible region is closed and bounoded, from the Weierstrass theorem it follows that C and D are global minima, as they minimize the objective function value.

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